Chemical Formulas

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	A subscript tells how many atoms of one element are in a formula unit, while a coefficient multiplies every atom in the entire formula.

Round 1 - Retrieve It

1. What does a subscript tell you?

2. What does a coefficient tell you?

3. In H2O, how many H atoms and O atoms are represented?

4. In 3H2O, how many water molecules are represented?

Round 2 - Sketch It

Use circles labeled H and O to draw one H2O molecule, then draw what 3H2O represents.

Chemical Formulas - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Count atoms in $3H_2O$.

2. Count atoms in $2CO_2$.

3. Count atoms in Al_2O_3.

Round 4 - Reason from Evidence

Compare $2H_2O$ and H_4O_2. They show the same total atom counts. Why do the symbols not mean the same thing chemically?

Chemical Formulas - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A **Articulate**
Explain Chemical Formulas in your own words. Use one example that is not printed on the cue card.

C **Connect**
Connect coefficients to conservation of atoms in chemical equations. Why do chemists adjust coefficients rather than subscripts when balancing?

E **Extend**
Write the total number of each atom in 4NH3 and explain your multiplication.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Chemical Formulas

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. The number of atoms of the element immediately before it.

2. How many whole molecules or formula units are present; it multiplies every atom in the formula.

3. 2 H and 1 O.

4. 3 molecules.

Sketch It - look for

One H_2O: two H atoms bonded with one O atom. $3H_2O$: three separate H_2O groups, totaling 6 H and 3 O atoms.

Apply It

1. 6 H and 3 O; 9 atoms total.

2. 2 C and 4 O; 6 atoms total.

3. 2 Al and 3 O; 5 atoms total.

Reason from Evidence - look for

A coefficient changes the number of intact H_2O molecules. Changing subscripts changes the formula/ratio within one substance and can represent a different substance.

ACE - Extend look-for

N = 4 and H = 12 because the coefficient 4 multiplies the entire NH_3 formula.

Suggested use	Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.